TUTORIAL

Diels–Alder reaction: a 2D energy surface

> Updated for EasyHybrid 3.2.0 (2026-10-26). The figures and numbers come from a run of this tutorial with the 3.2.0 development version.

In the Diels–Alder reaction a diene and an alkene (the dienophile) form a six-membered ring by making two new C–C bonds:

butadiene + ethylene → cyclohexene

Are the two bonds formed at the same time (a concerted, synchronous mechanism), one after the other (stepwise, through a diradical), or in between? A one-dimensional scan cannot answer it, since it fixes how the two bonds are combined. Scanning the **two** distances independently gives the energy surface over all their combinations, and its lowest path shows the mechanism.

In this tutorial you will:

1. build a starting complex with two **harmonic restraints**;
2. treat the whole system with PM6;
3. run a **2D scan** (14 × 14 points) on 12 processors;
4. use the 2D tools of the PES Analysis: stationary points and the lowest-barrier path.

The scan takes about 20 s on 12 cores. The structure is `examples/tutorials/03_diels_alder/diels_alder.mol2` in the EasyHybrid folder: butadiene and ethylene 3.6–3.8 Å apart.

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1. Create the system

File → New, **System Type** *DYFF*, the file `examples/tutorials/03_diels_alder/diels_alder.mol2`, name `diels_alder`. DYFF, the generic force field of EasyHybrid, types any organic molecule from its bonds; here it is used only for the first optimization. (The `.mol2` file has no charges, so the DYFF charges are zero: fine for this step.)

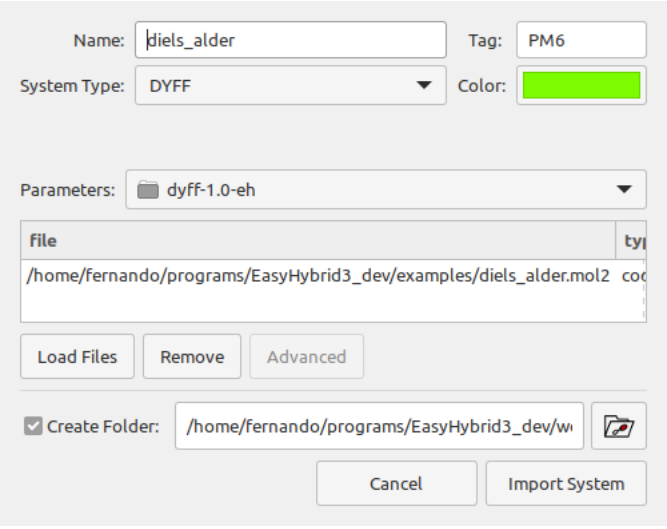


Figure 1. A DYFF system from a mol2 file.

2. Bring the molecules together with restraints

The two new bonds will be C8–C12 and C5–C11 (atom numbers of the mol2 file; indices 7–11 and 4–10 in EasyHybrid, which counts from 0). Bring the molecules to 2.8 Å with two harmonic restraints:

1. Switch to **Picking** mode and pick the first pair (C8, then C12).
2. Right-click in the 3D view and choose **Add Harmonic Restraint**. Set the **distance** to **2.8** Å and keep the force constant (4000 kJ mol^{-1} Å^{-2}). Click **Add**.
3. Do the same for C5 and C11.

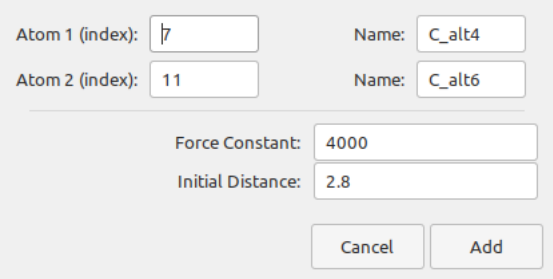


Figure 2. A harmonic restraint between the picked atoms.

The restraints are drawn as dashed lines and listed in **Selection → Selections and Restraints** (Figure 3), where each one can be switched off, edited or deleted.

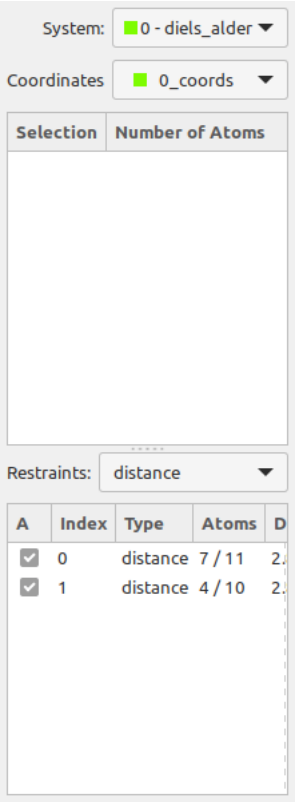


Figure 3. The two restraints of the system.

Optimize the geometry (**Simulate → Geometry Optimization**, RMS gradient 0.01). Active restraints enter every simulation of the system; the optimized structure has both distances at 2.81 Å (Figure 4). Then remove them: **System → Remove → Restraints**.

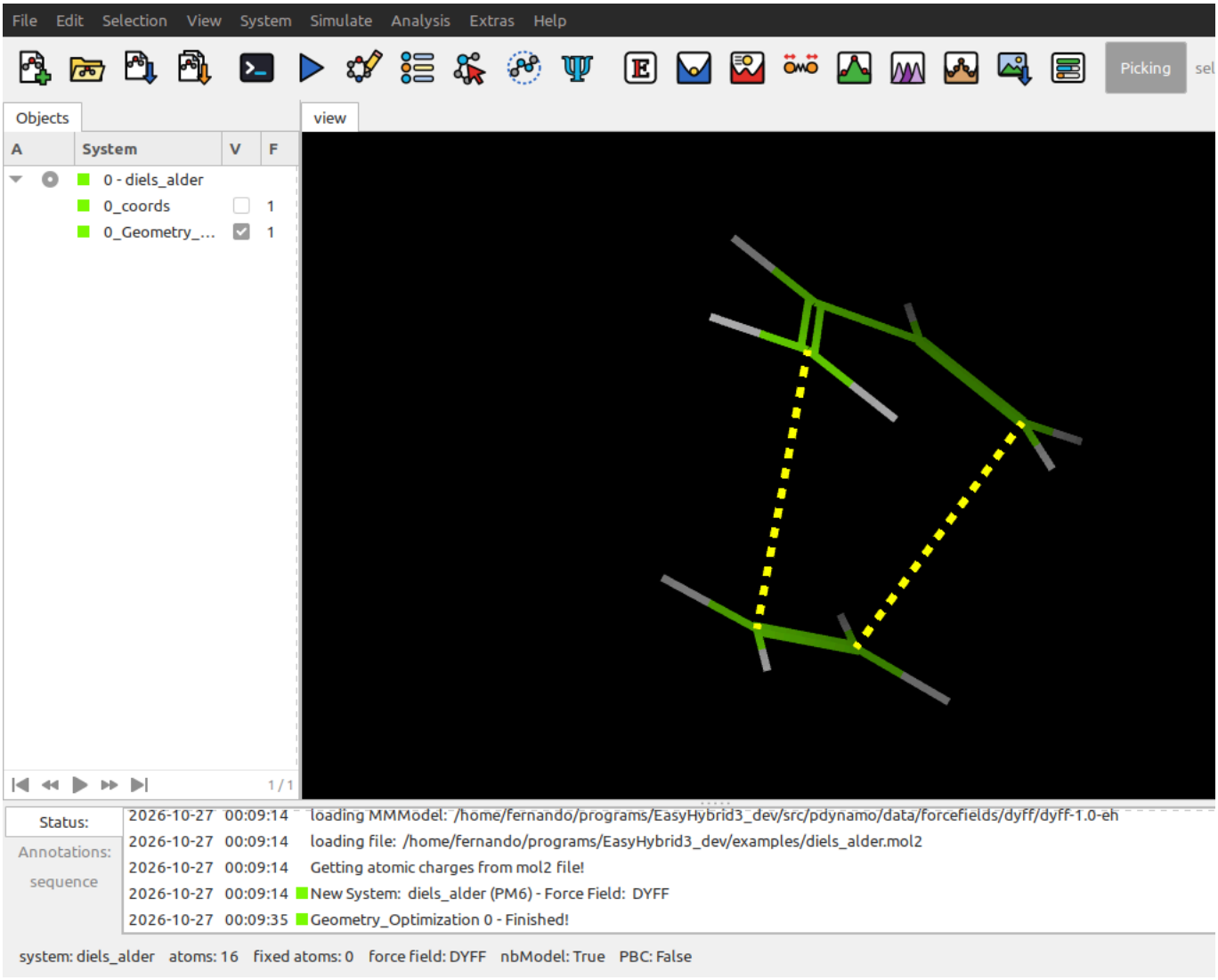


Figure 4. The complex after the restrained optimization.

3. The QC model

Press **Quantum Chemistry** in the toolbar, choose **pm6**, charge 0, multiplicity 1, and **OK**. With no atoms selected the QC model covers all 16 atoms: the force field is no longer used.

4. The 2D scan

Simulate → Reaction Coordinate Scans. For each coordinate, pick its two atoms and click **Import from Picking Selection**; tick **Reaction Coordinate 2** to enable the second one.

	Reaction Coordinate 1	Reaction Coordinate 2
Coordinate Type	simple distance	simple distance
Atoms	C8–C12 (7, 11)	C5–C11 (4, 10)
Initial Distance	2.8 Å	2.8 Å
Step Size	**−0.1** Å (the distance decreases)	−0.1 Å
Number of Steps	14 (2.8 → 1.5 Å)	14
Force Constant	4000	4000

Coordinates: the restrained optimized structure; **Method** Conjugate Gradient, **Maximum iterations** 1000, **RMS Grad. Tolerance** 0.1; **Number of CPUs** 12 (the columns of the grid are optimized in parallel); **Trajectory Name** `DA_scan2D` .

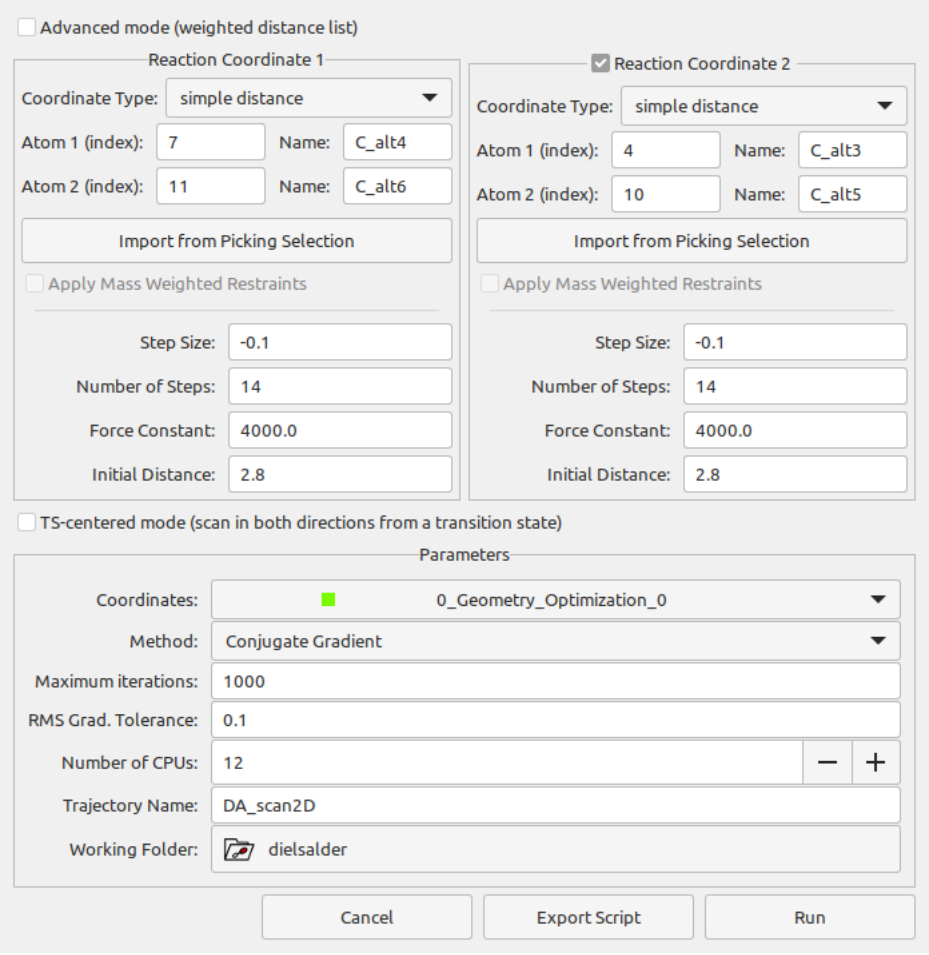


Figure 5. Two simple distances, scanned together.

Click **Run**. The 196 points are written as `frame<i>_<j>.pkl` in `DA_scan2D.ptGeo`. Import them from the *Finished!* dialog: the format is *pkl folder 2D*, chosen automatically.

5. The energy surface

Analysis → PES Analysis, with the 2D scan object. The map shows the energy (kJ/mol, relative to the lowest point) over RC1 (rows) and RC2 (columns); **Side by side** shows the map next to the profile of a path.

1. Tick **Mark minima and saddle points**: circles mark the minima, diamonds the saddle points.
2. Click **Lowest-barrier path**: EasyHybrid finds, between the two minima, the path whose highest point is the lowest, and plots its profile (Figure 6). You can also click points on the map to draw your own path.

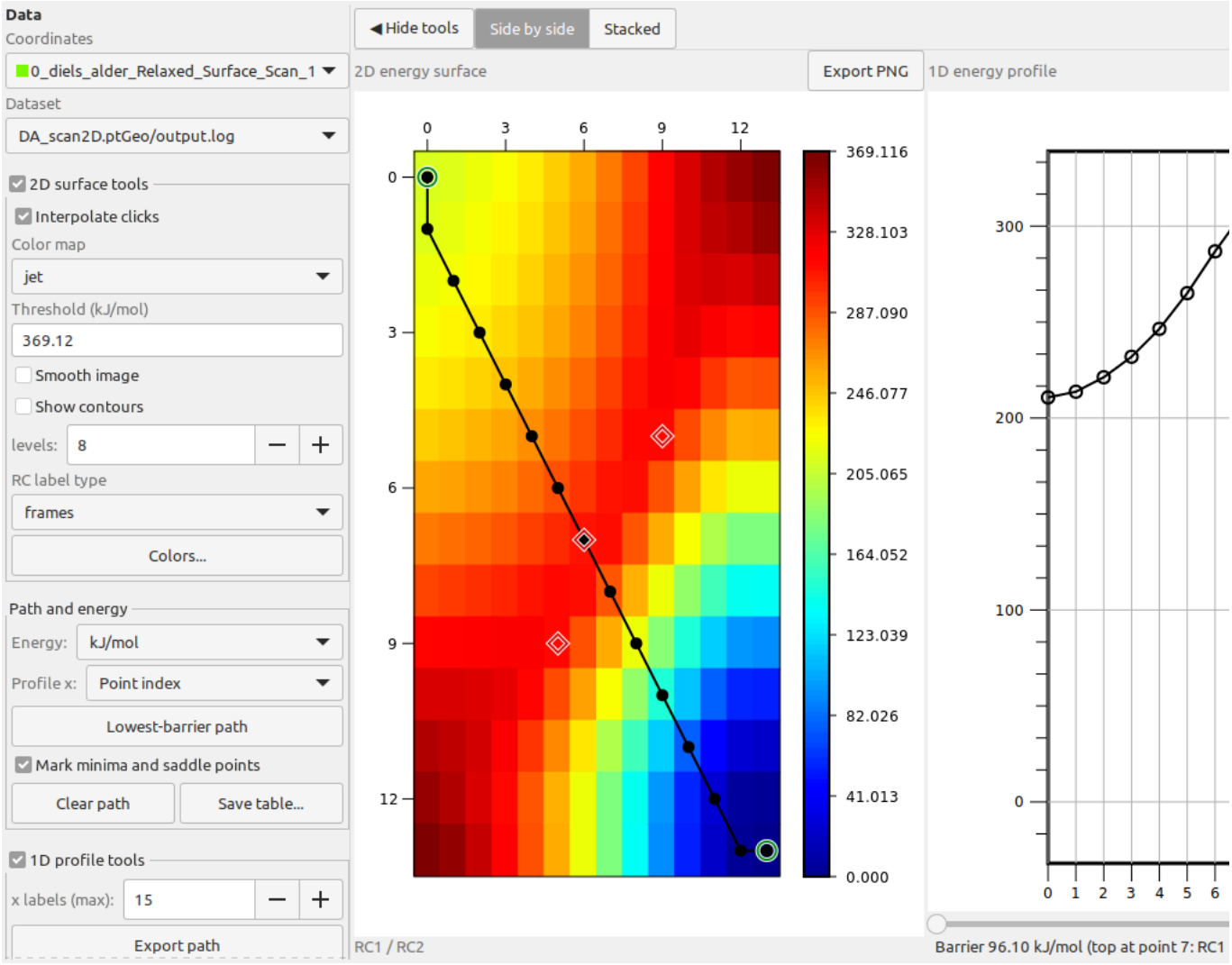


Figure 6. The PM6 surface. Top left: the reactant complex (2.8, 2.8 Å); bottom right: cyclohexene (1.5, 1.5 Å).

	RC1 (Å)	RC2 (Å)	E (kJ/mol)
Reactant complex	2.80	2.80	210.7
Lowest saddle point	2.11	2.21	306.8
Cyclohexene	1.50	1.50	0.0

- **Barrier**: 306.8 − 210.7 = **96 kJ/mol** (23 kcal/mol); the experimental activation energy of butadiene + ethylene is about 115 kJ/mol (27.5 kcal/mol).
- **Reaction energy**: −211 kJ/mol; PM6 overestimates it (experiment: about −170 kJ/mol).
- **Mechanism**: the lowest path runs close to the diagonal, with both bonds shortening together; at the saddle point they differ by only 0.1 Å (concerted and almost synchronous). The corners where one bond is formed and the other is still 2.8 Å apart (the stepwise, diradical route) are the highest regions of the map.

Click a cell of the map to show its structure in the 3D view.

Notes

- **Grid resolution.** The saddle point of a 0.1 Å grid is located within a grid cell. Refine it with **Simulate → Search Transition State (Baker)** from the structure of that cell.
- **Restraints stay active** until they are switched off or removed: forgetting them is a common cause of strange results in later simulations.
- **Reaction Coordinate 2 + Number of CPUs.** A 2D scan runs one column per process; 14 columns on 12 processors take about the time of two columns.

EasyHybrid 3.2.0 · built 2026-10-29 from `dev` @ `3539da4` · https://github.com/ferbachega/EasyHybrid3